$$m = -\left\{z + \frac{iac}{1-iam}\right\}^{-1} \quad (m = m(z; c)).$$

By solving this quadratic equation for $m(z; c)$ we obtain, by (1.8), the following expression for $\nu(\lambda; c)$:

$$\nu(\lambda; c) = \nu_1(\lambda; c) + \nu_2(\lambda; c),$$

where

$$\frac{d\nu_1(\lambda; c)}{d\lambda} = \begin{cases} (1-c)\delta(\lambda) & \text{for } 0 \leqslant c \leqslant 1, \\ 0 & \text{for } c > 1, \end{cases}$$

$$\frac{d\nu_2(\lambda; c)}{d\lambda} = \frac{\sqrt{\frac{1}{2}\left(\sqrt{(\lambda^2+\lambda_1^2)(\lambda^2+\lambda_2^2)} + \lambda^2 - \lambda_1\lambda_2\right) - \lambda}}{2a\lambda}.$$

$$\lambda_{1,2} = a(1 \pm \sqrt{c})^2.$$

From these formulas we see that in this case the spectrum occupies the entire axis, as was to be expected from the unboundedness of r.

We note in conclusion that it is, as a rule, impossible to find $m(z; c)$, and even more so to find $\nu(\lambda; c)$, in explicit form, since (1.14) is, generally speaking, not explicitly solvable for $m(z; c)$. In this sense the above examples are exceptional. Nevertheless, it is frequently possible to obtain a qualitative picture of the spectrum: the number and arrangement of its connected components, and also the behavior of $\nu(\lambda; c)$ near the boundary of the spectrum. We have in mind the following: on the intervals complementary to the spectrum on the real axis the function $m(x + i0; c)$ exists and is continuous, real and montonically increasing. Hence, the inverse function exists on these intervals—also real and monotonically increasing. Furthermore, its range of values is clearly just the complement of the spectrum. By denoting this inverse function by $x(m; c)$ and using (1.14) we obtain

$$x(m; c) = x_0(m; c) + c\int_{-\infty}^{\infty} \frac{\tau\, d\sigma(\tau)}{1+\tau m}, \tag{1.15}$$

where $x_0(m)$ is the function inverse to $m_0(x)$. Thus we have the following rule for determining the spectrum: it is necessary to locate the intervals where the function on the right-hand side of (1.15) is monotonically increasing and then to determine the set of its values on these intervals. The spectrum is the complement of this set.

Therefore if a is the endpoint of one of the intervals on which the right-hand side of (1.15) is monotonically increasing, then

$$\lambda_a = x_0(a) + c\int_{-\infty}^{\infty} \frac{\tau\, d\sigma(\tau)}{1+\tau a} \tag{1.16}$$

is the endpoint of one of the connected components of the spectrum. Suppose that in the neighborhood

From the strong law of large numbers we find that as $n \to \infty$ we almost certainly have $n^{-1}\sum_{i=1}^{n}\tilde{\tau}_i \to \int_{|x|\geq A}|x|d\sigma(x)$, whence, by the preceding inequality, Glivenko's theorem and the summability of $\tau(\xi)$, we conclude that, as $n \to \infty$, almost certainly

$$\int_0^{\beta_n} |\tau(\xi, T_n) - \tau(\xi)|\, d\xi \leqslant \frac{1}{A} + \int_{|x|\geqslant A} |x|\, d\sigma(x)$$

for any $A > 0$. But this means that $\int_0^{\beta_n}|\tau(\xi, T_n) - \tau(\xi)|d\xi$ almost certainly tends to zero for $n \to \infty$. Similarly, we find that $\int_{1-\beta_n}^{1}|\tau(\xi, T_n) - \tau(\xi)|d\xi$ almost certainly tends to zero as $n \to \infty$. Q.E.D.

Remark. This lemma is required below only for the case where the random quantity τ is bounded.

§ 3. Deduction of the basic equation

We shall prove Theorem 1 by first assuming the random quantities τ_i are bounded, i.e. there is a number $T > 0$ so that any realization of the τ_i satisfies

$$|\tau_i| < T. \tag{3.0}$$

This restriction will be lifted by going to a limit. But in the next two sections (3.0) is assumed to hold.

Suppose that T_n, Q_n is a realization of τ and q entering in (1.1). In this realization we number the pairs τ_i, $q^{(i)}$ in order of increasing τ_i:

$$\tau_1 \leqslant \tau_2 \leqslant \ldots \leqslant \tau_n, \quad q^{(1)} \leqslant q^{(2)} \leqslant \ldots \leqslant q^{(n)}$$

and construct a chain of operators $B_N(i)$ $(i = 1, 2, \cdots, n)$ by setting

$$B_N(i) = A_N + \sum_{\alpha=1}^{i} \tau_\alpha q^{(\alpha)}(\cdot, q^{(\alpha)}), \tag{3.1}$$

so that $B_N(0) = A_N$, and the $B_N(n)$ are the operators (1.1) that interest us. Denote the resolvents of the operators $B_N(i)$ by $R_z(i)$. For each realization T_n, Q_n we form the function $u(z, \xi; N, T_n, Q_n)$ defined for all nonreal z and real $\xi \in [0, 1]$ by

$$u(z, \xi; N, T_n, Q_n) = N^{-1}\operatorname{Sp} R_z(i) + nN^{-1}\operatorname{Sp}\{R_z(i+1) - R_z(i)\}\left(\xi - \frac{i}{n}\right)$$
$$\text{for} \quad \xi \in \left[\frac{i}{n}, \frac{i+1}{n}\right] \quad (i = 0, 1, \ldots, n-1). \tag{3.2}$$

Note that $u(z, 0; N, T_n, Q_n)$ and $u(z, 1; N, T_n, Q_n)$ are the Stieltjes transforms of the normalized spectral functions of the operators A_N and $B_N(n)$:

$$u(z, 0; N, T_n, Q_n) = N^{-1}\operatorname{Sp} R_z(0) = \int_{-\infty}^{\infty} \frac{d\nu(\lambda; A_N)}{\lambda - z}, \tag{3.3}$$

$$u(z, 1; N, T_n, Q_n) = N^{-1}\operatorname{Sp} R_z(n) = \int_{-\infty}^{\infty} \frac{d\nu(\lambda; B_N(n))}{\lambda - z}. \tag{3.3'}$$

For the remaining values of $\xi \in [0, 1]$ the function $u(z, \xi; N, T_n, Q_n)$ is the Stieltjes transform of relative to which $\nu(\lambda)$ is found by the inversion formula

$$\nu(\lambda_2) - \nu(\lambda_1) = \lim_{r \to 1-0} \frac{1}{2\pi} \int_{\lambda_1}^{\lambda_2} \operatorname{Re} n(re^{-i\theta})\, d\theta.$$

Theorem 3. *If as $N \to \infty$ the normalized spectral function of V_N tends to a function $\nu_0(\lambda)$ at all of its points of continuity and conditions* I *and* III *of* §1 *are satisfied, then the sequence of normalized spectral functions of the operators $U_N(n)$ converges in probability as $N \to \infty$ to a function $\nu(\lambda; c)$ at all of its points of continuity. The function $n(z; c)$ corresponding to $\nu(\lambda; c)$ by* (5.6) *is the solution to the equation*

$$\frac{\partial u(z, t)}{\partial t} + 2z \frac{\partial}{\partial z} \ln \left[1 + iu(z, t) \operatorname{tg} \frac{\tau(t)}{2} \right], \quad u(z, 0) = n_0(z) = \int_0^{2\pi} \frac{e^{i\lambda} + z}{e^{i\lambda} - z}\, d\nu_0(\lambda).$$

evaluated at $t = 1$. This equation has a unique solution (in the class of functions with positive real part for $|z| < 1$), given implicitly by the formula

$$u(z, t) = n_0\left(z \exp\left\{ -2c \int_0^t \frac{i \operatorname{tg} \frac{\tau(\xi)}{2}}{1 + iu(z, t) \operatorname{tg} \frac{\tau(\xi)}{2}}\, d\xi \right\} \right).$$

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